Name

Mathematics Advanced Year 11

# Piecewise-Defined Functions

**Book 1**

Version: 261021

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# Syllabus Content

**MAV-11-02** uses functions and relations to model, analyse and solve problems

**Piecewise-defined functions**

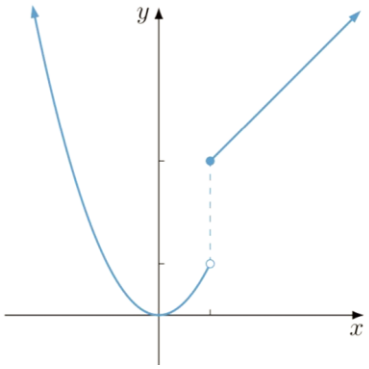
- Interpret piecewise-defined functions, where the function is defined differently in different parts of the domain
- Graph piecewise-defined functions involving functions covered in the scope of the Mathematics Advanced course, test if they are even or odd, and determine the domain and range
- Define informally that a function is continuous at a point if the curve can be drawn through the point without lifting the pen off the paper
- Identify points where piecewise-defined functions and other functions are not continuous
- Define a discontinuity of a function informally as a point where the function is not continuous

# Piecewise-Defined Function

## Piecewise-Defined Function

- A **piecewise function** combines more than one function in different parts of its domain.

$$f(x) = \begin{cases} g(x), & x \le a \\ h(x), & x > a \end{cases}$$




- $f(x)$ is called the piecewise-defined function.
- $g(x)$ and $h(x)$ are called subfunctions.
- To sketch a piecewise-defined function:
  1. Identify the graphs of each subfunction and their domains.
  2. Calculate function values at the boundaries.
  3. Sketch each subfunction separately.
     Decide whether to connect the curves or use open/closed circles.

## Example Sketch a piecewise-defined function

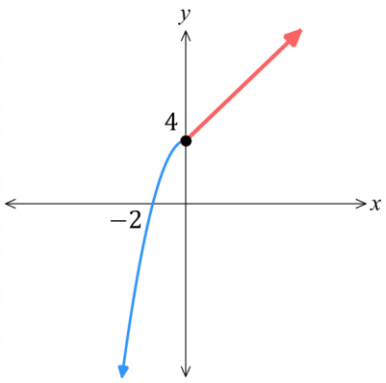
$$f(x) = \begin{cases} 4 - x^2, & x \le 0 \\ 4 + x, & x > 0 \end{cases}$$

1. Let $g(x) = 4 - x^2$ and $h(x) = 4 + x$

   $g(x)$: concave down parabola, vertex $(0,4)$

   $h(x)$: increasing straight-line.

2. $g(0) = 4$, $h(0) = 4$

3. 

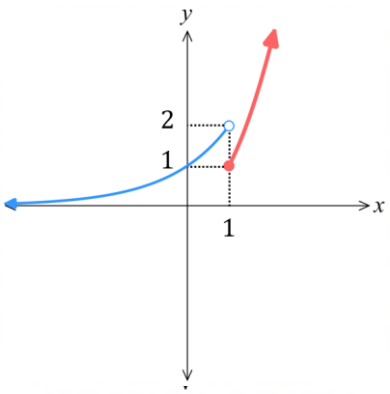

$$f(x) = \begin{cases} 2^x, & x < 1 \\ x^2, & x \ge 1 \end{cases}$$

1. Let $g(x) = 2^x$ and $h(x) = x^2$

   $g(x)$: exponential growth, intercept (0,1)

   $h(x)$: concave up parabola, vertex (0,0)

2. $g(1) = 2$, $h(1) = 1$

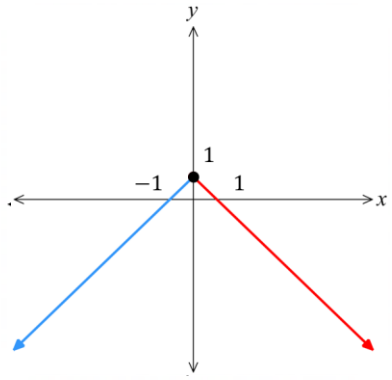
3. 

## Guided Practice

Sketch:

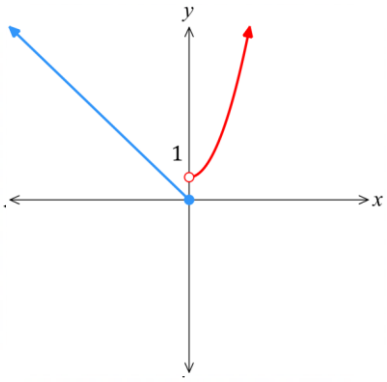
**a**

$$f(x) = \begin{cases} 1 + x, & x \leq 0 \\ 1 - x, & x > 0 \end{cases}$$



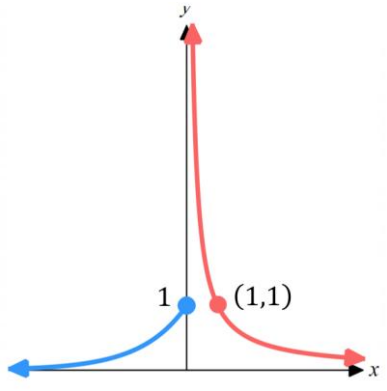

**b**

$$f(x) = \begin{cases} -x, & x \leq 0 \\ x^2 + 1, & x > 0 \end{cases}$$



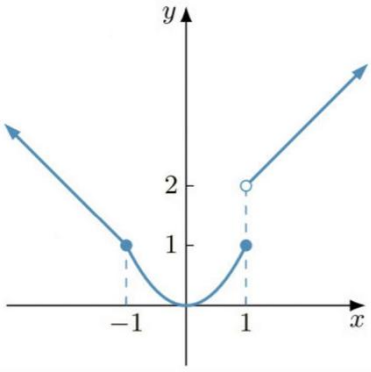

**c**

$$f(x) = \begin{cases} 2^x, & x \leq 0 \\ \dfrac{1}{x}, & x > 0 \end{cases}$$




**d**

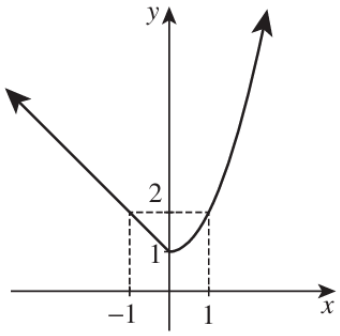
$$f(x) = \begin{cases} -x, & x < -1 \\ x^2, & -1 \leq x < 1 \\ x + 1 & x > 1 \end{cases}$$



**1** Sketch the following piecewise-defined functions.

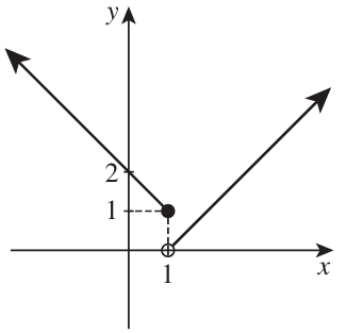
**a**

$$f(x) = \begin{cases} 1 - x, & \text{for } x < 0, \\ 1 + x^2, & \text{for } x \geq 0. \end{cases}$$



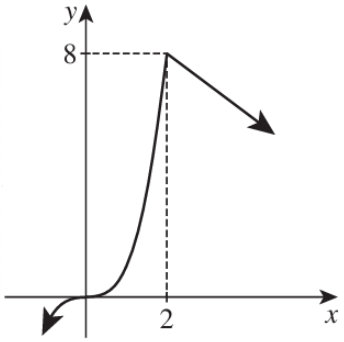

**b**

$$y = \begin{cases} 2 - x, & \text{for } x \leq 1, \\ x - 1, & \text{for } x > 1. \end{cases}$$



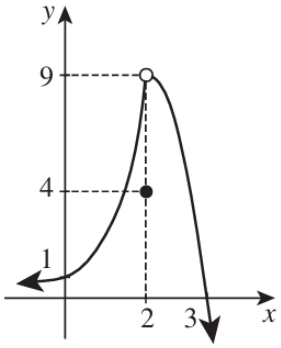

**c**

$$f(x) = \begin{cases} x^3, & \text{for } x \leq 2, \\ 10 - x, & \text{for } x > 2. \end{cases}$$




**d**

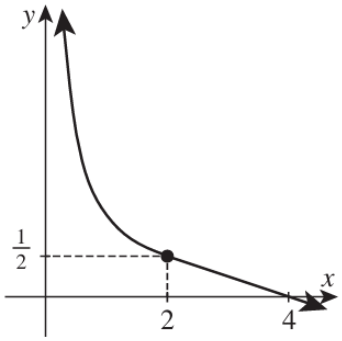
$$f(x) = \begin{cases} 3^x, & \text{for } x < 2, \\ 13 - x^2, & \text{for } x > 2, \\ 4, & \text{for } x = 2. \end{cases}$$




**2** Sketch these piecewise functions.

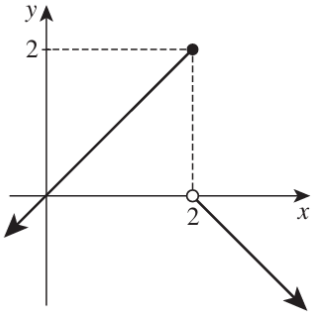
**a**

$$f(x) = \begin{cases} \frac{1}{x}, & \text{for } 0 < x < 2, \\ 1 - \frac{1}{4}x, & \text{for } x > 2, \\ \frac{1}{2}, & \text{for } x = 2. \end{cases}$$




**b**

$$f(x) = \begin{cases} x, & \text{for } x < 2, \\ 2 - x, & \text{for } x > 2, \\ 2, & \text{for } x = 2. \end{cases}$$



**3** Consider the piecewise-defined function:

$$f(x) = \begin{cases} 2-x, & \text{for} \quad x < -2 \\ 4, & \text{for} \quad -2 \leq x \leq 2 \\ x+2, & \text{for} \quad x > 2 \end{cases}$$

Evaluate:

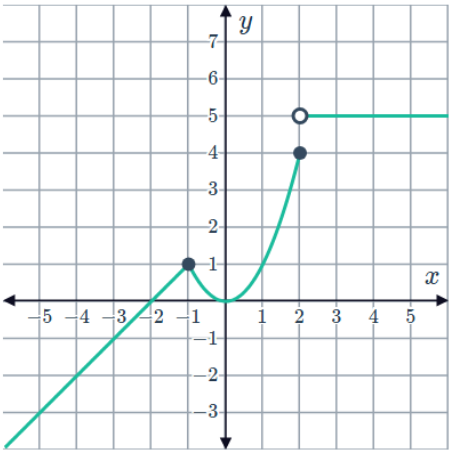
**a** $f(0)$ 4

**b** $f(5)$ 7

**c** $f(-3)$ 5

**d** $f(3) - f(-1)$ 1

**4** Consider the piecewise defined function and evaluate:




**a** $f(-1)$ 1

**b** $f(4)$ 5

**c** $f(2)$ 4

**d** $f(-4) - f(3)$ −7

**5** Consider the piecewise defined function and evaluate:

$$f(x) = \begin{cases} x^2 - 2; & x \geq 3 \\ 4; & 0 < x < 3 \\ 4x; & x \leq 0 \end{cases}$$

**a** $f(4)$ 14

**b** $f(0.5)$ 4

**c** $f(-3)$ −12

**d** $f(a)$, where $a$ is negative $4a$

**6** Consider the piecewise defined function and evaluate:

$$f(x) = \begin{cases} 4x - 3; & x \geq 0 \\ 7 - x^2; & x < 0 \end{cases}$$

**a** $f(5) - f(-5)$ 35

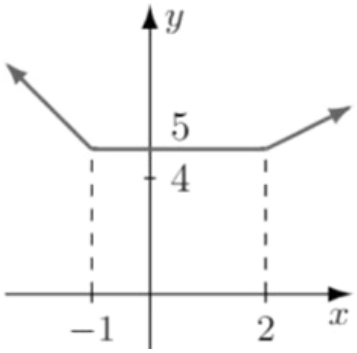
**b** $6f(6)$ 126

**c** $f(a^2)$ $4a^2 - 3$

**7** Sketch these piecewise-defined functions and state their range.

**a**

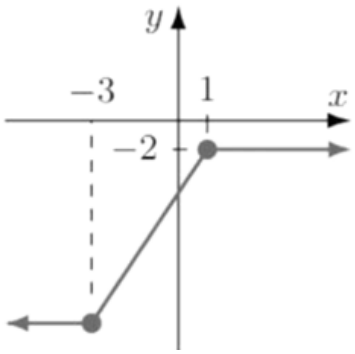
$$f(x) = \begin{cases} -2x+3, & \text{if } x < -1 \\ 5, & \text{if } -1 \le x \le 2 \\ x+3, & \text{if } x > 2 \end{cases}$$




$R: [5, \infty)$

**b**

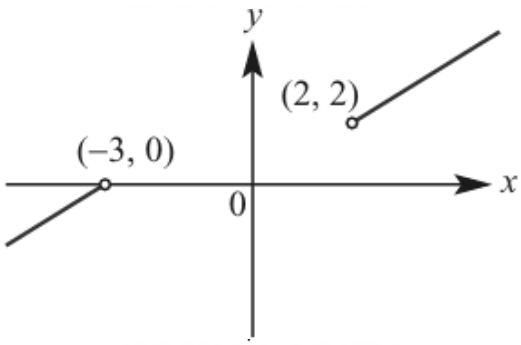
$$f(x) = \begin{cases} 3, & \text{if } x < -3 \\ 3x-5, & \text{if } -3 \le x \le 1 \\ -2, & \text{if } x > 1 \end{cases}$$




$R: (-\infty, -2]$

**c**

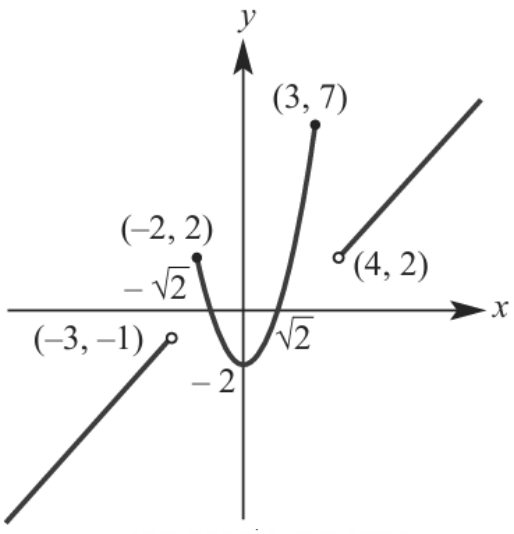
$$f(x) = \begin{cases} x+3, & x < -3 \\ x, & x > 2 \end{cases}$$




$R: (-\infty, 0) \cup (2, \infty)$

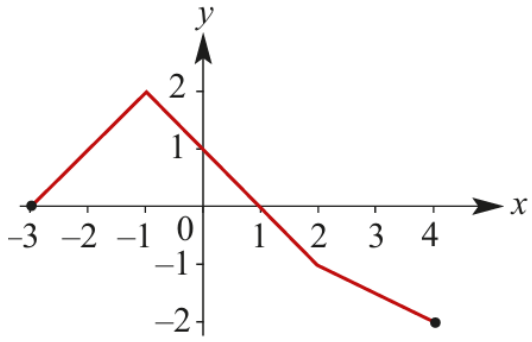
**d**

$$f(x) = \begin{cases} x+2, & x < -3 \\ x^2-2, & -2 \le x \le 3 \\ x-2, & x > 4 \end{cases}$$




$R: (-\infty, -2]$

**8** Specify the function represented by this graph:



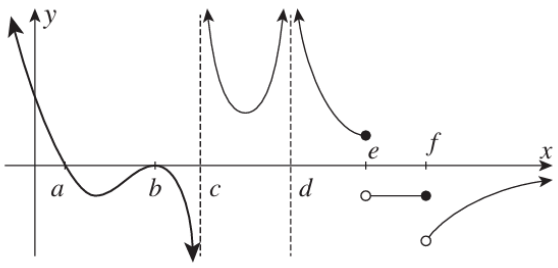

$$f(x) = \begin{cases} x+3, & -3 \le x \le -1 \\ -x+1, & -1 < x \le 2 \\ -\frac{x}{2}, & 2 < x \le 4 \end{cases}$$

# Continuous at a Point

## ⓘ Continuous at a Point

A function is **continuous** at a point $x = a$ if the graph of $y = f(x)$ can be drawn through the point in a single pen stroke without lifting the pen.

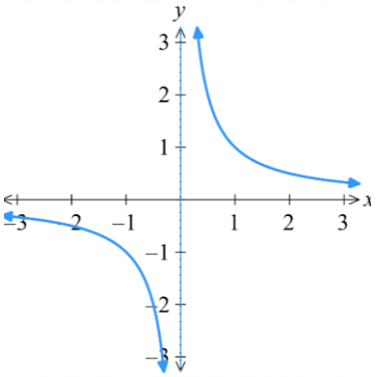
In the graph below, the function is continuous everywhere except for the **discontinuities** at $x \in \{c, d, e, f\}$.



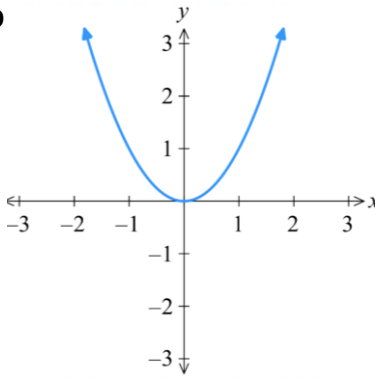

## 🔍 **Identify** discontinuities in graphs.

Identify any discontinuities in these curves, if any.

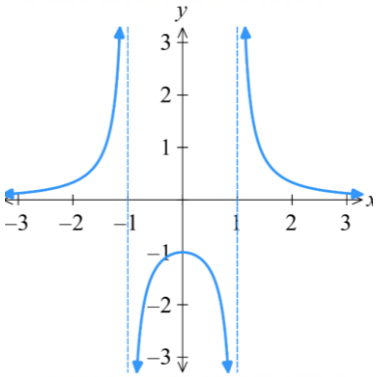
**a**



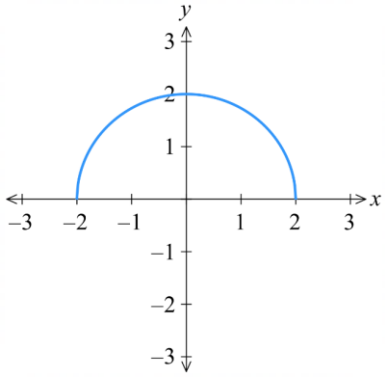

**b**



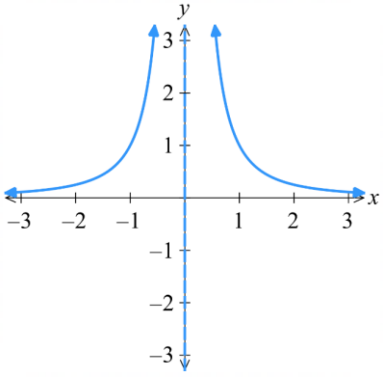

**c**



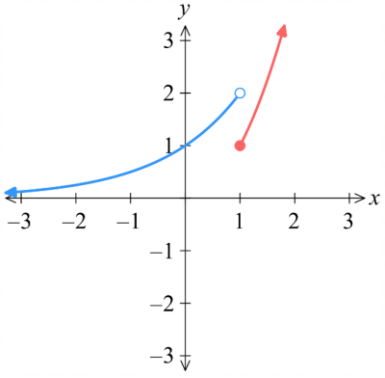

**d**




**e**




**f**



## Continuous from an Equation

A function is **discontinuous** at $x = a$ if:

- The function is undefined at that point (e.g. asymptote, denominator = 0).
- A piecewise function doesn't join up at $x = a$.

Given a piecewise function:

$$f(x) = \begin{cases} g(x), & x \leq a \\ h(x), & x > a \end{cases}$$

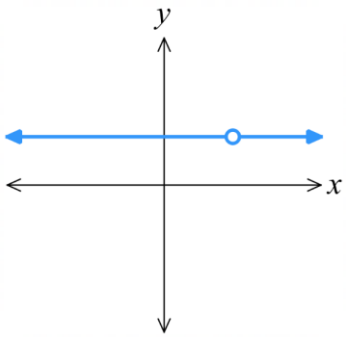
The function is **continuous** at $x = a$ if $g(a) = h(a)$.
The function is **discontinuous** at $x = a$ if $g(a) \neq h(a)$.

We will assume the curve is continuous everywhere else.

## Example Determine continuous from an equation.

Decide whether these functions are continuous at $x = 1$ using the equation. A sketch is given.

$f(x) = \dfrac{3x - 3}{4x - 4}$

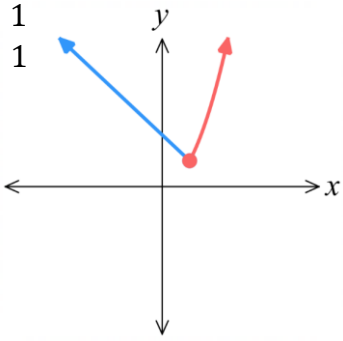



$f(x)$ is undefined when $4x - 4 = 0$

$x = 1$

Since $f(x)$ is undefined at $x = 1$, it is not continuous at $x = 1$.

$f(x) = \begin{cases} 2 - x, & x < 1 \\ x^2, & x \geq 1 \end{cases}$




Let $g(x) = 2 - x$

$g(1) = 1$

Let $h(x) = x^2$

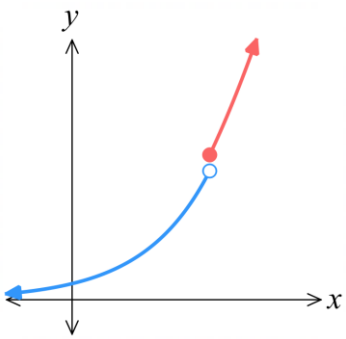
$h(1) = 1$

The graphs join up at $x = 1$, $\therefore$ continuous.

## Practice

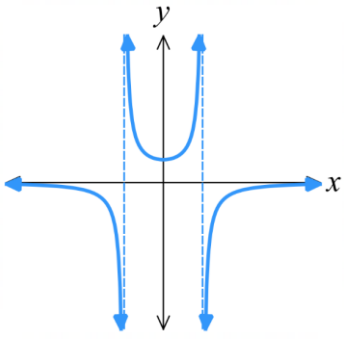
Decide whether these functions are continuous at $x = 3$.

$f(x) = \begin{cases} 2^x, & x < 3 \\ x^2, & x \geq 3 \end{cases}$




Discontinuity at $x = 3$

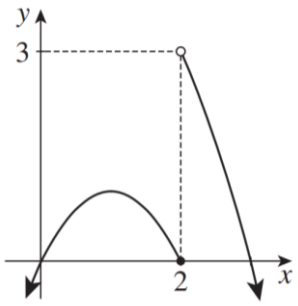
$f(x) = \dfrac{5}{x^2 - 4}$




Continuous at $x = 3$

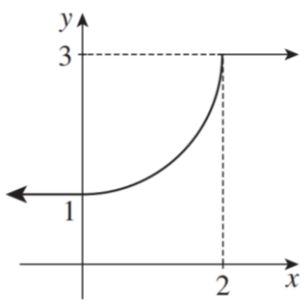
**1** State whether each function is continuous at the given point.

**a** $x = 2$



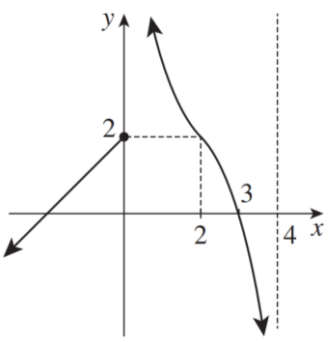

Discontinuous

**b** $x = 2$



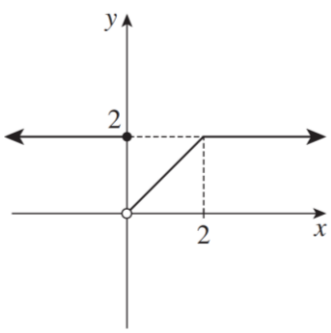

Continuous

**c** $x = 0$




Discontinuous

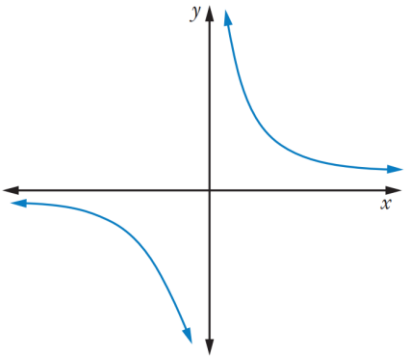
**d** $x = 2$




Continuous

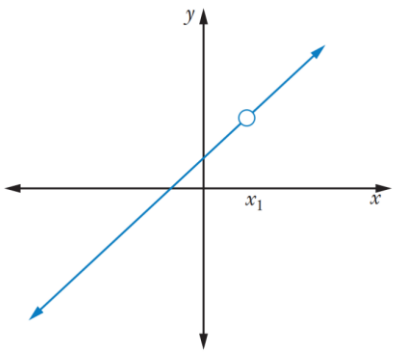
**2** For these functions, state whether the function is continuous throughout its domain. If not, identify the $x$ value(s) where the function has a discontinuity.

**a**



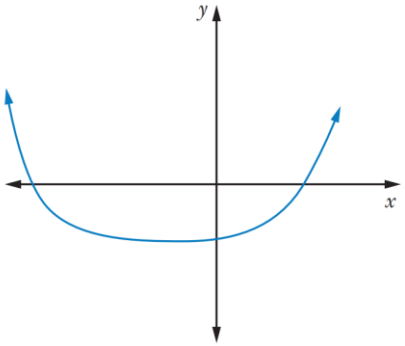

discontinuity at $x = 0$

**b**



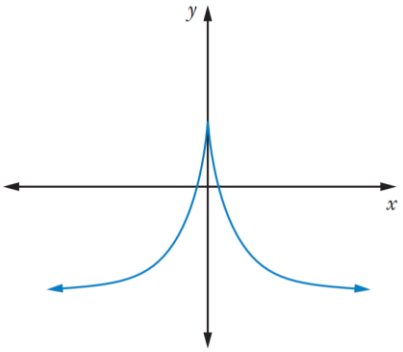

discontinuity at $x = x_1$

**c**



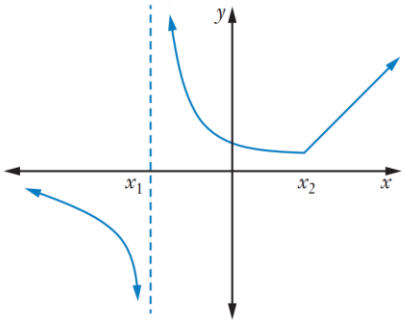

continuous function

**d**



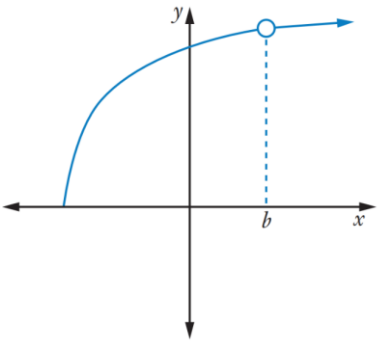

continuous function

**e**




discontinuity at $x = x_1$

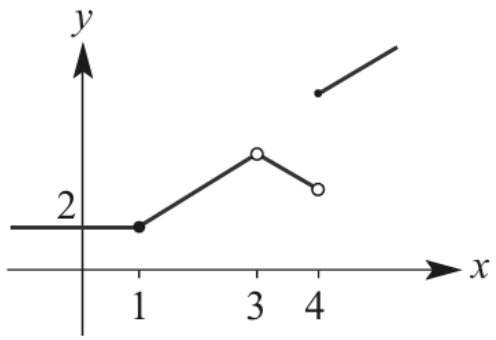
**f**




discontinuity at $x = b$

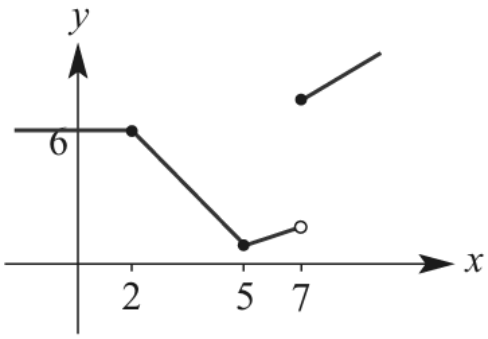
**3** State the values of $x$ at which a discontinuity occurs.

**a**



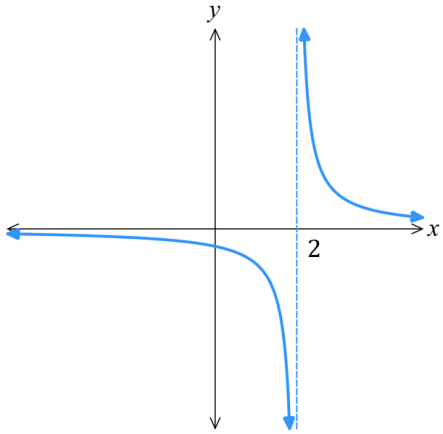

$x = 3$ and $4$

**b**




$x = 7$

**c**




$x = 2$

**d**

$$f(x) = \begin{cases} 3x & \text{if } x \geq 0 \\ -2x + 2 & \text{if } x < 0 \end{cases}$$

$x = 0$

**e**

$$f(x) = \begin{cases} -x & \text{if } x \leq -1 \\ x^2 & \text{if } -1 < x < 0 \\ -3x + 1 & \text{if } x \geq 0 \end{cases}$$

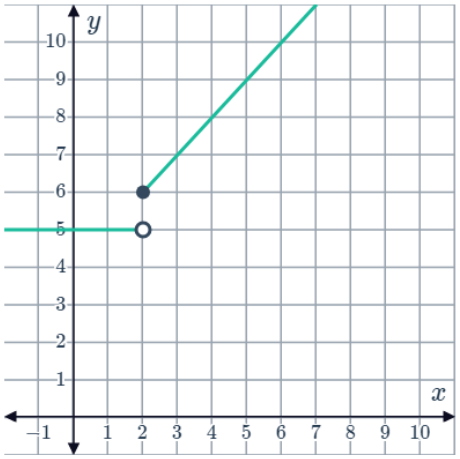
x = 0

$$y = \begin{cases} 2 & \text{if } x < 1 \\ (x-4)^2 - 9 & \text{if } 1 \leq x < 7 \\ x - 7 & \text{if } x \geq 7 \end{cases}$$

$x = 1$

**4** Consider the graph of $y = f(x)$.

**a**



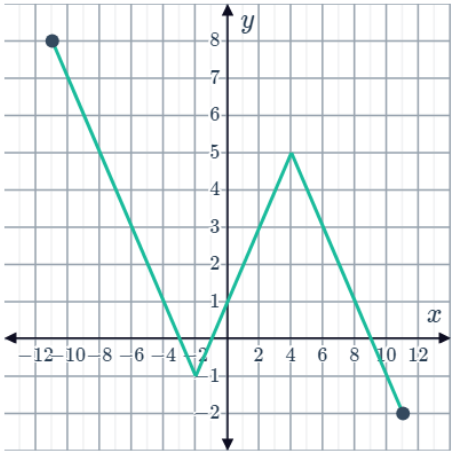

**i.** What is the function value at $x = 2$?

6

**ii.** Is the graph continuous at $x = 2$?

no

**b**




**i.** What is the function value at $x = 4$?

5

**ii.** Is the graph continuous at $x = 4$?

yes

**5** Determine where the function is discontinuous, using your knowledge of asymptotes and/or natural domains. Check your answer using DESMOS.

**a**

$$f(x) = \frac{5}{6 - x}$$

$x = 6$

**b**

$$f(x) = \frac{3x}{(x-1)(x-3)(x-5)}$$

$x = 1, 3, 5$

**c**

$$f(x) = \frac{x(x+1)}{(x+1)(x+3)}$$

$x = -1, -3$

**d**

$$f(x) = \frac{5}{x^2 - 5x}$$

$x = 0, 5$

**e**

$$f(x) = \frac{x}{x^2 - 5x + 6}$$

$x = 3, 2$

**f**

$$f(x) = \frac{x^2 - 16}{x^2 - 9}$$

$x = -3, 3$